Luke William Johnston

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Diabetes epidemiologist, R user and developer, teacher of modern data science and analysis skills, and open science and reproducibility advocate. *Only showing most items **since 2019**. For full CV, see cv.lukewjohnston.com.*

Education	
2013 – 2017	PhD in Nutritional Sciences, University of Toronto in Toronto, CA
2011 – 2013	MSc in Nutritional Sciences, University of Toronto in Toronto, CA
2005 – 2010	BSc in Kinesiology, University of Waterloo in Waterloo, CA

Positions held	
2019 – present	DDA postdoctoral researcher, Steno Diabetes Center Aarhus in Aarhus, Denmark

Teaching and curriculum planning and development

Teaching experience.	
2020 – present	Lead instructor for Reproducible Research in R: An intermediate workshop on modern approaches and workflows to processing data (PhD and Postdocs), Danish Diabetes Academy in Denmark
2020 – present	Lead instructor for Reproducible Research in R: An introductory workshop on modern data analyses and workflows (PhD and Postdocs), Danish Diabetes Academy in Denmark
2019	Lead instructor for Software Carpentry Workshop (Anyone), Aarhus University in Aarhus, Denmark
2019	Lead instructor for Reproducible Quantitative Analyses and Workflows using R (PhD and Postdocs), German Institute of Human Nutrition Potsdam-Rehbrueke in Potsdam, Germany
2019	Lead instructor for Reproducible Quantitative Methods: Data analysis workflow using R (PhD and Postdocs), Danish Diabetes Academy in Denmark

Course planning.	
2020 – 2021	Scientific program organizing committee for the Danish Diabetes Academy Postdoctoral Winter School, Danish Diabetes Academy in Denmark

Curriculum development

1. 'rostools/r-cubed: Reproducible Research in R - An introductory workshop on modern data analyses and workflows'. **Luke W. Johnston**, Helene Bæk Juel, Bettina Lengger, Daniel R. Witte. (2020). URL: https://r-cubed.rostools.org/. DOI: 10.5281/zenodo.3921894.
2. 'r-cubed: Reproducible Research in R - An Intermediate Workshop on Modern Approaches and Workflows to Processing Data'. **Luke Johnston**, Omar Silverman-Retana, Andreas Eiset, Signe Storgaard. (2020). URL: https://r-cubed-intermediate.rostools.org/. DOI: 10.5281/zenodo.4061900.
3. 'Reproducible Quantitative Methods: Data analysis workflow using R'. **Luke W Johnston**, Daniel R Witte, João Santiago, Anna Schritz. (2019). URL: https://dda-rcourse.lwjohnst.com/. DOI: 10.5281/zenodo.2603311.
4. 'acdcourse: Analyzing Cohort Datasets with R'. **Luke W Johnston**. (2019). URL: https://github.com/lwjohnst86/acdcourse. DOI: 10.5281/zenodo.3247819.
5. 'UofTCoders/rcourse: Quantitative Methods in R for Biology'. **Luke W Johnston**, Madeleine Bonsma-Fisher, Joel Ostblom, Ahmed Hasan, James Santangelo, Lindsay Coome, Lina Tran, Elliott Sales De Andrade, Sara Mahallati. (2019). URL: https://uoftcoders.github.io/rcourse/. DOI: 10.5281/zenodo.1117432.

Awards and external funding

Showing all awards, excluding travel grants, during PhD and postdoc.	
2019	Danish Diabetes Academy Postdoctoral Fellowship from Danish Diabetes Academy (DKK 1.8 million), Denmark
2017	Michael C. Archer Research Excellence Award from University of Toronto , Toronto, Canada
2017	Gordon Cressy Student Leadership Award from University of Toronto , Toronto, Canada
2014	New Investigator Award from International Society for the Study of Fatty Acids and Lipids , Stockholm, Sweden
2014	Doctoral Student Research Award from Canadian Diabetes Association (CAD 21000), Toronto, Canada
2014	Margaret and Nicholas Fodor Fellowship from University of Toronto (CAD 3000), Toronto, Canada

	Funding obtained.
2022	A framework for an open and scalable infrastructure for health data exemplified by the DD2 initiative (NNF21OC0069462) from NovoNordiskFonden (DKK 14,977,580), Aarhus, Denmark

Dissemination activities

Publications since 2019

1. 'Dihydroceramide- and ceramide-profiling provides insights into human cardiometabolic disease etiology'. C. Wittenbecher, R. Cuadrat, **L. Johnston**, F. Eichelmann, S. Jäger, O. Kuxhaus, M. Prada, F. Del Greco M., A. A. Hicks, P. Hoffman, J. Krumsiek, F. B. Hu, M. B. Schulze. (2022). DOI: 10.1038/s41467-022-28496-1.
2. 'Escaping irreproducible research practices and spreading awareness through education and (re-)training'. Bettina Lengger, **Luke W. Johnston**. (2022). DOI: 10.1042/bio_2022_103.
3. 'Metabolomic profiling of the Dietary Approaches to Stop Hypertension diet provides novel insights for the nutritional epidemiology of type 2 diabetes mellitus'. Shahen Yashpal, Angela D. Liese, Beatrice A. Boucher, Lynne E. Wagenknecht, Steven M. Haffner, **Luke W. Johnston**, Richard P. Bazinet, Marian Rewers, Jerome I. Rotter, Steve M. Watkins, Anthony J. Hanley. (2021). DOI: 10.1017/s0007114521003561.
4. 'r-cubed: Guiding the overwhelmed scientist from random wrangling to Reproducible Research in R'. **Luke Johnston**, Helene Juel, Bettina Lengger, Daniel Witte, Hannah Chatwin, Malene Christiansen, Anders Isaksen. (2021). DOI: 10.21105/jose.00122.
5. 'Next generation diabetes scientists shape global research culture'. M. M. Broadley, A. Gonzalez-Franquesa, A. Jonsson, C. B. Christiansen, G. D. Carrasquilla, A. Mamidi, S. M. Ghiasi, H. B. Juel, S. Falk, M. S. Isidor, P. Aldiss, L. Gillberg, K. Carolo dos Santos, R. Sabaratnam, L. O. Huang, J. S. Quist, J. R. Knudsen, S. Poulsen, R. Quaranta, M. Tozzi, R. B. Mikkelsen, M. K. Andersen, M. Dall, A. B. Møller, M. H. Drag, S. Panahi, C. L. Lyons, L. Small, A. Altıntaş, P. Poursharifi, B. Osborne, A. K. Sarvari, **L. W. Johnston**, M. H. Solheim, I. M. Modvig, A. S. Husted, N. Z. Jespersen, E. L. Brown, E. Bak, A. Peluso, F. Finger, K. V. Grunddal, K. Rupar, H. T. Vistisen, J. B. Henningsen, T. Ma, M. Hauge Pedersen, K. Plucinska, S. L. Jepsen, G. Repasky. (2020). DOI: 10.1111/apha.13455.
6. 'Effect of familial diabetes status and age at diagnosis on type 2 diabetes risk: a nation-wide register-based study from Denmark'. Omar Silverman-Retana, Adam Hulman, Jannie Nielsen, Claus T. Ekstrøm, Bendix Carstensen, Rebecca K. Simmons, Lasse Bjerg, **Luke W. Johnston**, Daniel R. Witte. (2020). DOI: 10.1007/s00125-020-05113-8.
7. 'A graduate student-led participatory live-coding quantitative methods course in R: Experiences on initiating, developing, and teaching'. **Luke Johnston**, Madeleine Bonsma-Fisher, Joel Ostblom, Ahmed Hasan, James Santangelo, Lindsay Coome, Lina Tran, Elliott De Andrade, Sara Mahallati. (2019). DOI: 10.21105/jose.00049.
8. 'The Distribution of Fatty Acid Biomarkers of Dairy Intake across Serum Lipid Fractions: The Prospective Metabolism and Islet Cell Evaluation (PROMISE) Cohort'. Ingrid D. Santaren, Richard P. Bazinet, Zhen Liu, **Luke W. Johnston**, John L. Sievenpiper, Adria Giacca, Ravi Retnakaran, Stewart B. Harris, Bernard Zinman, Anthony J. Hanley. (2019). DOI: 10.1002/lipd.12185.
9. 'The macrophage activation marker soluble CD163 is longitudinally associated with insulin sensitivity and β-cell function'. Zhila Semnani-Azad, Philip W Connelly, **Luke W Johnston**, Ravi Retnakaran, Stewart B Harris, Bernard Zinman, Anthony J Hanley. (2019). DOI: 10.1210/clinem/dgz166.
10. 'Associations of Calcium from Food Sources versus Phosphate Binders with Serum Calcium and FGF23 in Hemodialysis Patients'. Sara Mahdavi, Antonio Bellasi, Karan Nagra, **Luke Johnston**, Paul Tam, Biagio Di Iorio, Tabo Sikaneta. (2019). DOI: 10.3390/jcm8101680.
11. 'Determinants of longitudinal change in insulin clearance: the Prospective Metabolism and Islet Cell Evaluation cohort'. Zhila Semnani-azad, **Luke W Johnston**, Christine Lee, Ravi Retnakaran, Philip W Connelly, Stewart B Harris, Bernard Zinman, Anthony J Hanley. (2019). DOI: 10.1136/bmjdrc-2019-000825.

Books

1. 'Research Software Engineering with Python'. Damien Irving, Kate Hertweck, **Luke Johnston**, Joel Ostblom, Charlotte Wickham. (2021). URL: https://merely-useful.tech/py-rse/. ISBN: 0367698323.

R packages

	R Packages
2022	NetCoupler: Inference of Causal Links Between a Network and an External Variable (creator), CRAN
2014	broom: Convert Statistical Objects into Tidy Tibbles (contributor), CRAN
2016	prodigenr: Research Project Directory Generator (creator), CRAN
2016	mason: Build Data Structures for Common Statistical Analysis (creator), CRAN
2019	posterdown: Generate PDF Conference Posters Using R Markdown (contributor), CRAN
2016	carpenter: Build Common Tables of Summary Statistics for Reports (creator), CRAN

Presentations (since 2020)

1. (Oral) 'NetCoupler: Inferring causal pathways between high-dimensional metabolic data and external factors'. **Luke W. Johnston**, Clemens Wittenbecher, Fabian Eichelmann. *Annual useR Conference* in Virtual (2021).

2. (Contributed) 'r-cubed: Guiding the overwhelmed scientist from random wrangling to Reproducible Research in R'. Hannah Chatwin, **Luke W Johnston**, Helene Bæk Juel, Bettina Lengger, Daniel R. Witte, Malene Revsbech Christiansen, Anders Aasted Isaksen. *Annual useR Conference* in Virtual (2021).
3. (Oral) 'NetCoupler: Inferring causal pathways between high-dimensional metabolic data and external factors'. **Luke W. Johnston**, Clemens Wittenbecher. *Annual Meeting of the Danish Epidemiological Society* in Virtual (2021).
4. (Oral) 'How I Teach Life Scientists About Reproducibility and Data Analysis Using R'. **Luke W. Johnston**. *Life Science Trainers Talk Series "How I Teach Life Scientists..."* in Virtual (2021).
5. (Oral poster) 'The metabolic pathways between components of stature and HbA (1c): a causal structure learning approach in the UK Biobank'. **LW Johnston**, C Wittenbecher, HT Vistisen, CC Dahm, DR Witte. *European Association for the Study of Diabetes Annual Meeting* in virtual setting (2020).
6. (Oral) 'NetCoupler: Inferring causal pathways between high-dimensional metabolic data and external factors'. **L. W. Johnston**, C. Wittenbecher. *IARC Nutritional Methodology and Biostatistics Group* in Virtual (2020).